

Agent365 Sampling V4 Summary Report

STATUS Validated hash, schema, and selection-stage linkage	GENERATED AT 2026-08-05T13:52:26.307194Z
POPULATION 2,800 sessions	RUNTIME 3h 11m 27s

Executive Summary

This report answers one question: under fixed exact token budgets, which sampling method best approximates census outcome rates while preserving coverage. The current run covers 2,800 sessions and completed in 3h 11m 27s. Percent tier labels are provenance labels only; exact token budgets are the authoritative comparison axis.

Experiment Design Three methods compared at matched exact budgets across 5 budget levels with persisted memberships and deterministic fills.	Outcome Winner Pattern Random won 4 of 5 budgets; MinHash won 1 of 5 budgets. Exact values are in the Outcomes tab.
IDW Verdict Embedding IDW improved 6 cells, regressed 9, tied 0.	MinHash Verdict 4,310 bucket misses were treated as novel; 0 exhaustive scans were used.

Representative Exact-Budget Table

Representative budget uses 20% legacy provenance when available; otherwise median exact budget.

Reported method	Exact token budget	Selected sessions mean	Reported MAE	Fraction saved	Concept coverage
Random selected-only	263,797	566.00	0.0130	79.8%	48.3%
MinHash selected-only	263,797	508.33	0.0252	81.8%	50.5%
Full-session embedding + IDW	263,797	369.33	0.0402	86.8%	65.1%

Metric Definitions

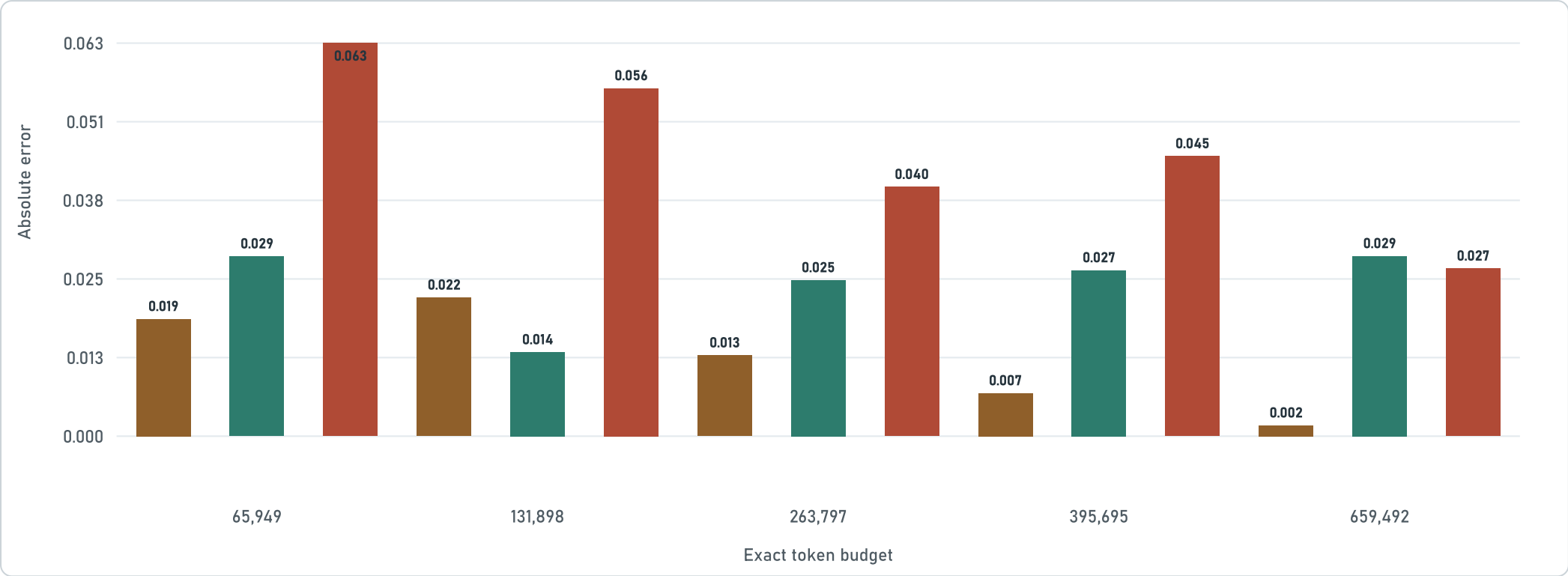
- Selected-only absolute aggregate rate error = $|\text{selected pass rate} - \text{census}|$.
- IDW aggregate error = $|\text{observed+imputed population mean} - \text{census}|$.
- Fraction saved = $1 - (\text{selected count} / \text{population count})$.
- Concept coverage = $\text{covered concepts} / \text{total concepts in the run aggregate}$.
- Deterministic expected labels are pseudo-judge outputs; adaptive methods are not design-unbiased estimators.

Outcomes

Outcome comparisons keep exact budget parity. Random and MinHash report selected-only error; full-session embedding reports the observed-plus-IDW population estimate.

Mean absolute error versus census by exact token budget

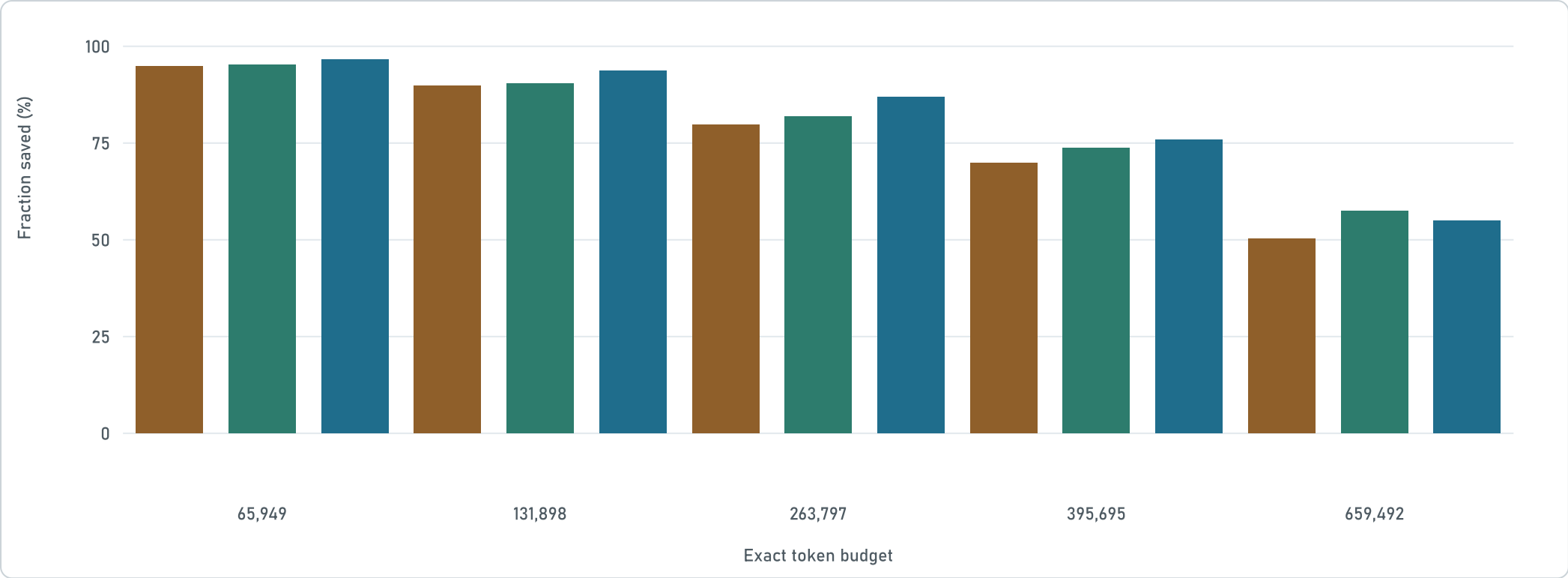
Random selected-only MinHash selected-only Full-session embedding + IDW



Interpretation: 65,949: Random selected-only lowest MAE 0.0189; 131,898: MinHash selected-only lowest MAE 0.0135; 263,797: Random selected-only lowest MAE 0.0130; 395,695: Random selected-only lowest MAE 0.0069; 659,492: Random selected-only lowest MAE 0.0017.

Fraction saved by exact token budget

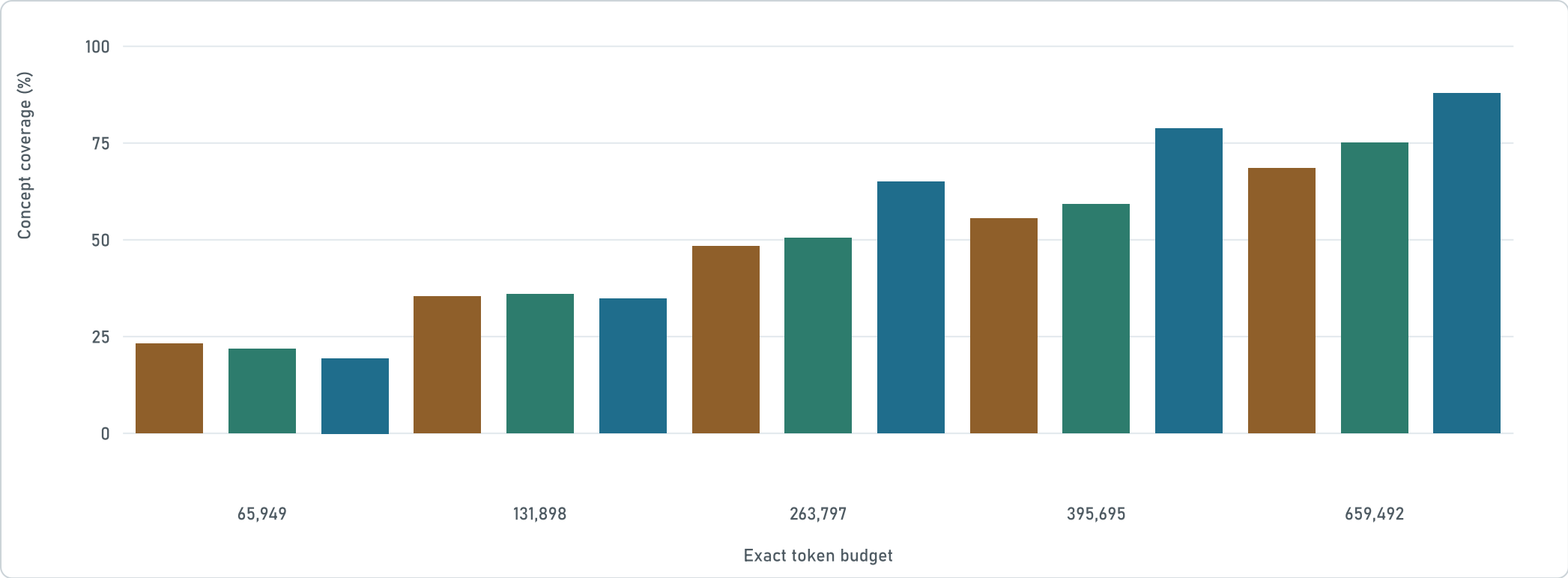
Random selected-only MinHash selected-only Full-session embedding + IDW



Highest fraction saved: 65,949: Full-session embedding + IDW 96.5%; 131,898: Full-session embedding + IDW 93.7%; 263,797: Full-session embedding + IDW 86.8%; 395,695: Full-session embedding + IDW 75.8%; 659,492: MinHash selected-only 57.5%.

Concept coverage by exact token budget

Random selected-only MinHash selected-only Full-session embedding + IDW



Highest concept coverage: 65,949: Random selected-only 23.2%; 131,898: MinHash selected-only 35.9%; 263,797: Full-session embedding + IDW 65.1%; 395,695: Full-session embedding + IDW 78.7%; 659,492: Full-session embedding + IDW 87.9%.

Dense Outcome Table

Legacy provenance tier	Exact tokens	Reported method	Selected count mean	Reported MAE	Fraction saved	Concept coverage	Note
5%	65,949	Random selected-only	147.33	0.0189	94.7%	23.2%	best reported MAE
5%	65,949	MinHash selected-only	131.67	0.0290	95.3%	21.8%	
5%	65,949	Full-session embedding + IDW	97.00	0.0634	96.5%	19.4%	
10%	131,898	Random selected-only	287.00	0.0224	89.8%	35.4%	best reported MAE
10%	131,898	MinHash selected-only	268.33	0.0135	90.4%	35.9%	
10%	131,898	Full-session embedding + IDW	176.33	0.0561	93.7%	34.8%	
20%	263,797	Random selected-only	566.00	0.0130	79.8%	48.3%	best reported MAE

Legacy provenance tier	Exact tokens	Reported method	Selected count mean	Reported MAE	Fraction saved	Concept coverage	Note
20%	263,797	MinHash selected-only	508.33	0.0252	81.8%	50.5%	
20%	263,797	Full-session embedding + IDW	369.33	0.0402	86.8%	65.1%	
30%	395,695	Random selected-only	843.67	0.0069	69.9%	55.4%	best reported MAE
30%	395,695	MinHash selected-only	738.67	0.0268	73.6%	59.1%	
30%	395,695	Full-session embedding + IDW	677.00	0.0452	75.8%	78.7%	
50%	659,492	Random selected-only	1393.67	0.0017	50.2%	68.5%	best reported MAE
50%	659,492	MinHash selected-only	1191.33	0.0290	57.5%	75.0%	
50%	659,492	Full-session embedding + IDW	1263.00	0.0270	54.9%	87.9%	

Sampling Methods

The complete sampling and estimation workflow is defined below.

Random token-priority selection

- 1. Build an exact token budget from persisted budget manifests.
- 2. Rank sessions deterministically from the paired randomized replay order.
- 3. Pack maximal whole sessions without splitting.
- 4. Seal labels until membership is frozen.

Estimation mode	Selected-only
Budget rule	Exact token budget is authoritative
Fill behavior	No separate adaptive fill stage

MinHash LSH 32x4 selection

- 1. Generate a 128-value MinHash signature split into 32 bands of 4 rows.
- 2. Propose sessions from candidate collisions under exact budget.
- 3. If no candidates collide, mark proposal as novel/no-candidate.
- 4. Do not execute exhaustive full-corpus fallback scans.

LSH shape	128 values; 32 bands x 4 rows
No-candidate handling	Novel proposal; no exhaustive scan
Estimation mode	Selected-only

Embedding selection

- 1. Embed sessions under the same exact budget process.
- 2. Select native embedding proposals with deterministic budget fill.
- 3. Freeze membership before any outcome estimation.
- 4. Track observed coverage and unjudged units for IDW stage.

Estimation mode	Selected-only baseline + IDW extension
Label handling	Labels remain sealed until membership freeze
Budget packing	Maximal whole-session packing

Same-agent IDW estimation

- 1. After membership freeze, impute only unjudged units.
- 2. Use same-agent donor neighborhoods with inverse-distance weighting.
- 3. Fallback to prior only when no donor is available.
- 4. Combine observed and imputed means into aggregate rate estimate.

k	8
power	2.000
distance epsilon	0.00000100
prior probability	0.500